April 1st 2026

Physics Concerto Seminar Series
Supported by the Weinberg Institute for Theoretical Physics,
Department of Physics, University of Texas

Chasing Shadows:

A Glimpse into the Hunt for Dark Matter

Gabriele Montefalcone

Weinberg Institute for Theoretical Physics, University of Texas at Austin

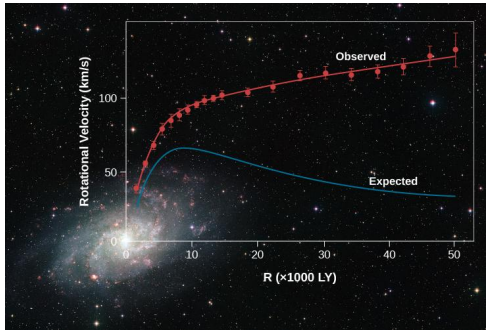
TEXAS
The University of Texas at Austin

PHYSICS CONCERTO
BRIDGING PHYSICS SPECIALTIES THROUGH PEER TO PEER SEMINARS

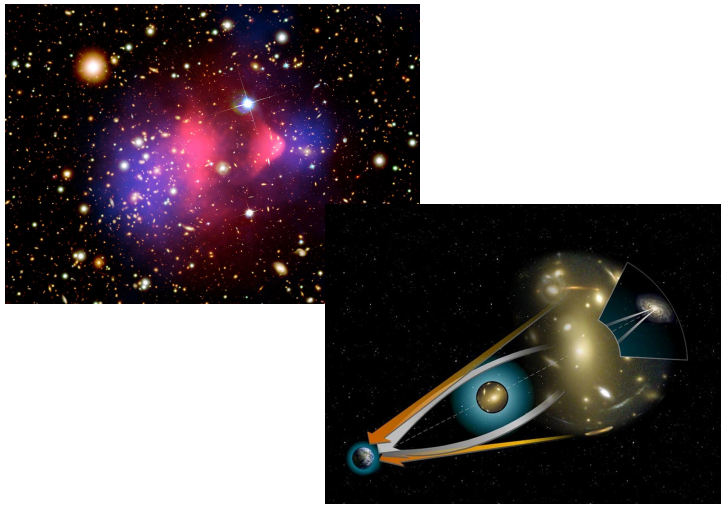
Evidence for Dark Matter (DM)

Huge amount of evidence from all scales (only from gravitational interaction)

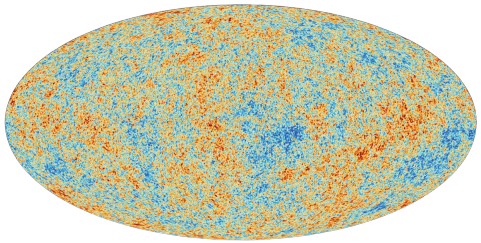
Galactic scales

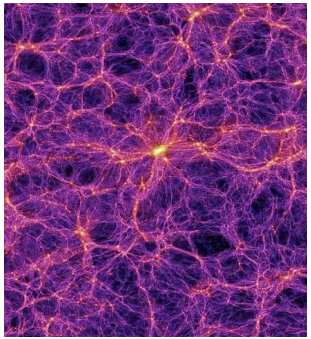


Cluster scales



Cosmological scales





What we know about DM

- Cold and Massive
- Stable/long lived
- No/weak interactions with the Standard Model (SM)
- No/weak SM charge (electric and color)
- Abundance: DM corresponds to **%25** of the energy budget in the universe today (**~5x** the amount of ordinary matter)

$$\Omega_c h^2 \approx 0.12$$

What we know about DM

- Cold and Massive
- Stable/long lived
- No/weak interactions with the Standard Model (SM)
- No/weak SM charge (electric and color)
- Abundance: DM corresponds to **%25** of the energy budget in the universe today (**~5x** the amount of ordinary matter)

$$\Omega_c h^2 \approx 0.12$$

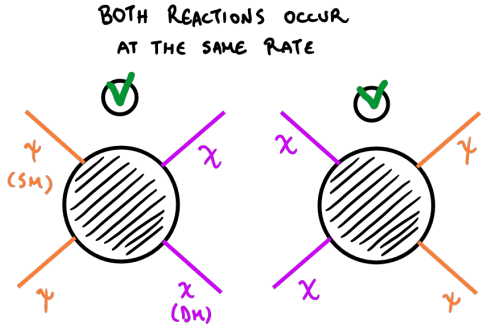
How was DM produced in the early universe?

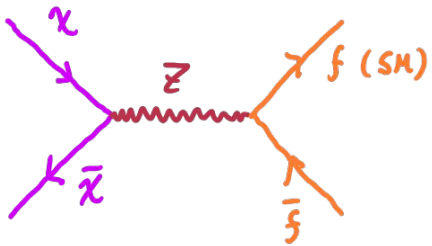
The Simplest DM
The Canonical Freeze-out Story

Add a heavy fermion **χ** to the SM, coupled to the **Z**

$$\mathcal{L} \supset g_\chi \, Z_\mu \, \bar{\chi}\gamma^\mu\chi \,, \qquad g_\chi \sim \mathcal{O}(g_W)$$

- electrically neutral, massive (**m_{DM} ~ GeV-TeV**)
- in thermal equilibrium with SM when **$T \gg m_{DM}$**



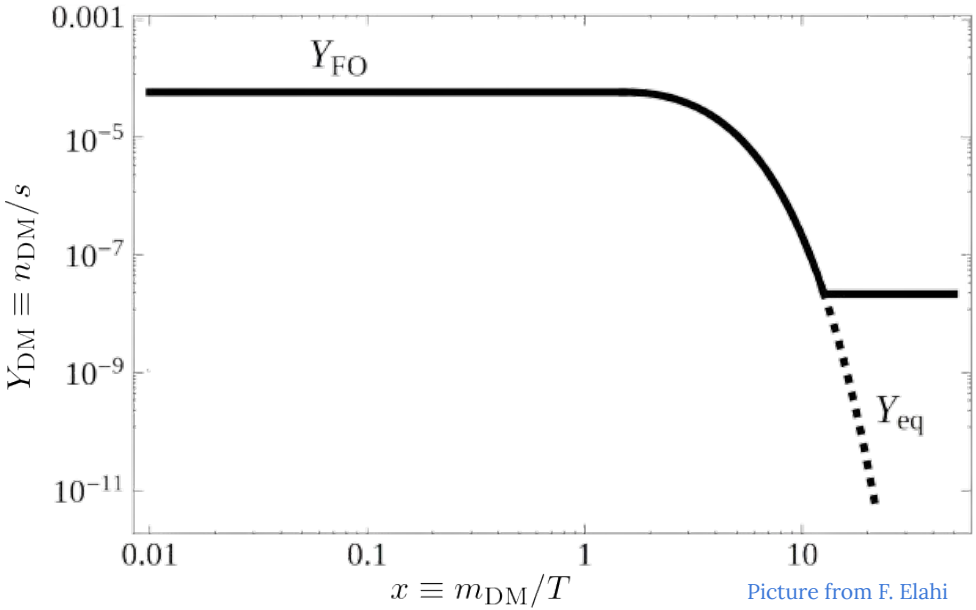


The Simplest DM

The Canonical Freeze-out Story

$$\mathcal{L} \supset g_\chi \, Z_\mu \, \bar{\chi}\gamma^\mu\chi \,, \qquad g_\chi \sim \mathcal{O}(g_W)$$

- DM freezes out at $T \simeq m_{DM}/20$



Picture from F. Elahi

THE WIMP MIRACLE

$m_{DM} \simeq m_W$ and $\sigma_{DM} \simeq \alpha_W^2/m_W^2$
reproduces the observed DM abundance
($\alpha_W \simeq 10^{-2}$, $m_W \simeq 100$ GeV)

$$\langle\sigma v\rangle \sim \frac{g_W^4}{m_\chi^2} \sim \frac{\alpha_W^2}{m_\chi^2} \sim \frac{(0.01)^2}{(100\,\mathrm{GeV})^2} \sim 10^{-27}\,\mathrm{cm}^3\,\mathrm{s}^{-1}$$

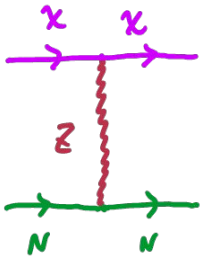
$$\Omega_c h^2 \approx \frac{10^{-26}\,\mathrm{cm}^3\,\mathrm{s}^{-1}}{\langle\sigma v\rangle}$$

Target

$\Omega_c h^2 \approx 0.12$

The Simplest DM
Cracks in the WIMP

$$\mathcal{L} \supset g_\chi Z_\mu \bar{\chi}\gamma^\mu \chi\,, \qquad g_\chi \sim \mathcal{O}(g_W)$$

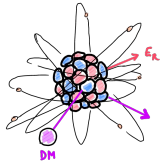


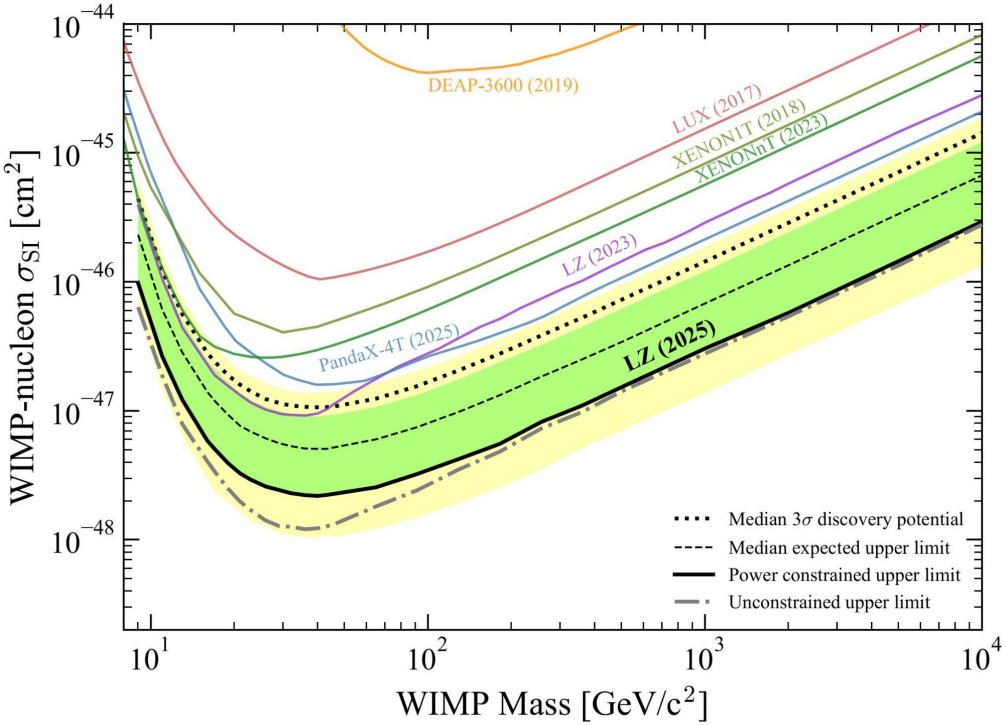
$$\sigma^{\rm SI}_{\chi N} \sim \frac{G_F^2\, m_N^2}{\pi} \sim 10^{-39}\ {\rm cm}^2$$

The simplest WIMPs are ruled out by direct detection.

- We need a new mediator.

Nuclear Recoil:
GeV-TeV DM

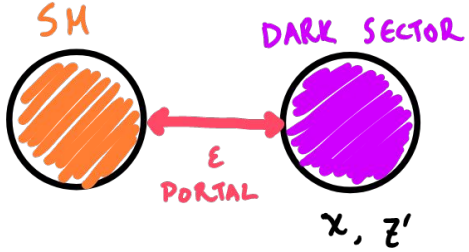




A Minimal Dark Matter Scenario
Mediating through a Dark Photon

Give DM its own gauge force: **U'(1),** with gauge boson **Z'**

$$\mathcal{L} \supset \underbrace{-\frac{\epsilon}{2} B^{\mu\nu} Z'_{\mu\nu}}_{\text{portal}} + \underbrace{g_D Z'_\mu \bar{\chi} \gamma^\mu \chi}_{\text{dark sector}}$$



- **Z'** couples to SM like a faint photon: $\varepsilon e Q_f$
- **χ** couples to **Z'** with dark gauge coupling g_D

m_χ	$m_{Z'}$	$\alpha_D = g_D^2/4\pi$	ε
DM mass	mediator mass	dark coupling	portal coupling

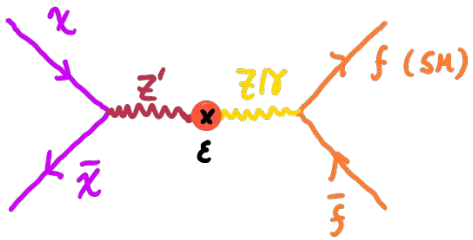
Four parameters → a zoo of cosmological histories

A Minimal Dark Matter Scenario
Freezing Out through a Dark Photon

$$m_\chi \gtrsim 1\,\mathrm{GeV}$$
$$m'_Z > 2\,m_\chi$$

$$\langle \sigma v \rangle \sim \frac{\alpha_{\mathrm{EM}}\,\alpha_D\,\epsilon^2\,m_\chi^2}{m_{Z'}^4}$$

Direct annihilation freeze-out



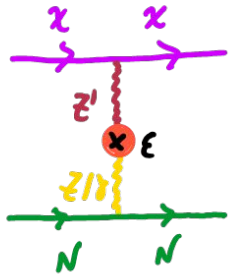
Small but not tiny α_D, tunable with ε and $R=m_{Z'}/m_{DM}$

$$\kappa \equiv \epsilon\sqrt{\alpha_D/\alpha_{\mathrm{EM}}} \sim \mathcal{O}(10^{-3} - 10^{-1})$$

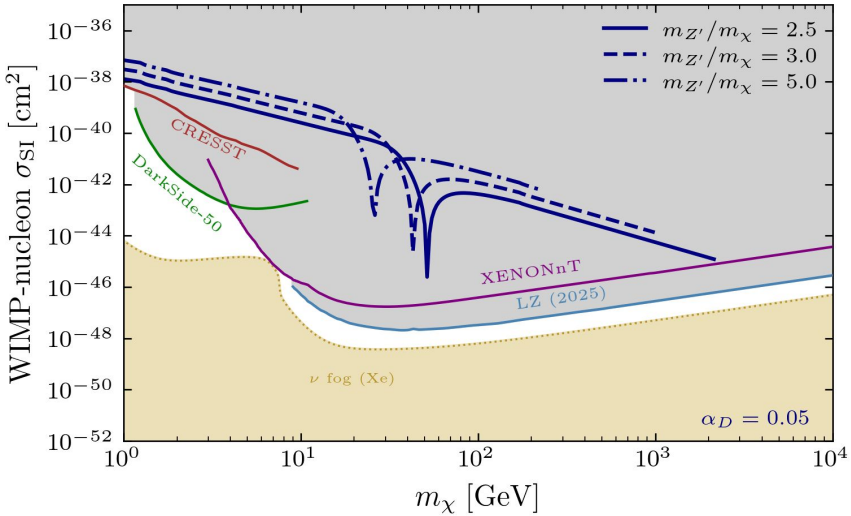
A Minimal Dark Matter Scenario
Freezing Out through a Dark Photon

$$m_\chi \gtrsim 1\,\mathrm{GeV}$$
$$m'_Z > 2\,m_\chi$$

$$\sigma^{\mathrm{SI}}_{\chi N} = \frac{16\pi\,\alpha_D\,\alpha_{\mathrm{EM}}\,\epsilon^2\,\mu^2_{\chi N}}{m^4_{Z'}}$$



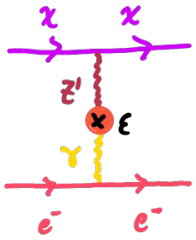
Effectively excluded by XENONnT and LZ 2025



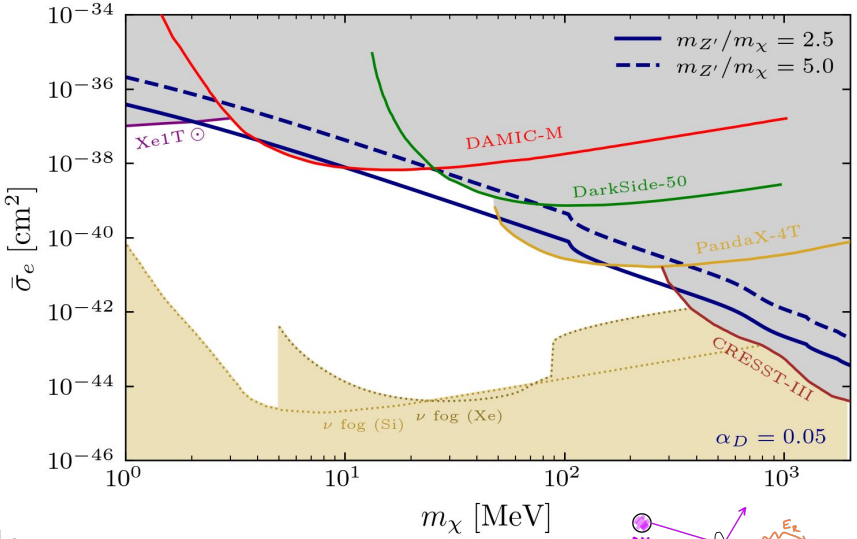
A Minimal Dark Matter Scenario
Freezing Out through a Dark Photon

$$m_\chi \lesssim 1\,\mathrm{GeV}$$
$$m'_Z > 2\,m_\chi$$

$$\bar{\sigma}_e = \frac{16\pi\ \alpha_{\mathrm{EM}}{}^2\ \mu_{\chi e}^2}{m_{Z'}^4}$$

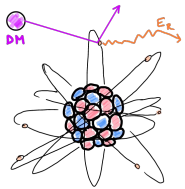


$$\kappa \equiv \epsilon\sqrt{\alpha_D/\alpha_{\mathrm{EM}}} \sim \mathcal{O}(10^{-4})$$



Only small regions of the parameter space remain viable

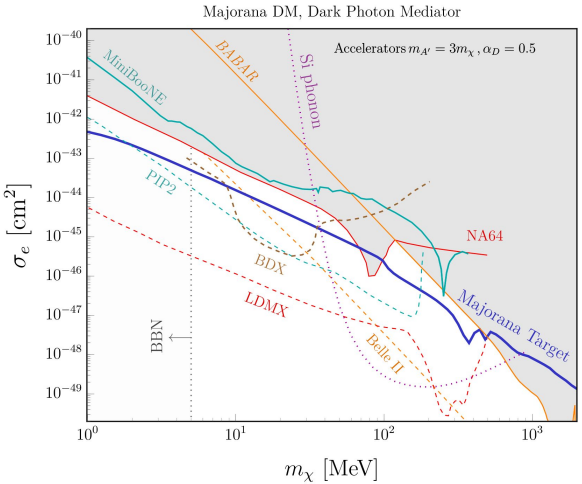
Electron Recoil:
sub-GeV DM



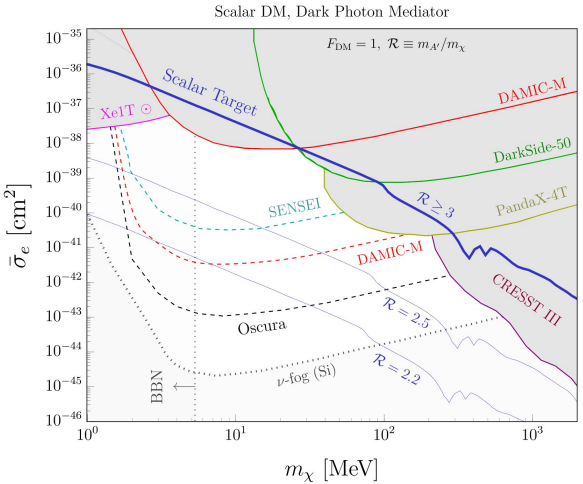
A Minimal Dark Matter Scenario
Freezing Out through a Dark Photon

$$m_\chi \lesssim 1\,\text{GeV}$$
$$m'_Z > 2\,m_\chi$$

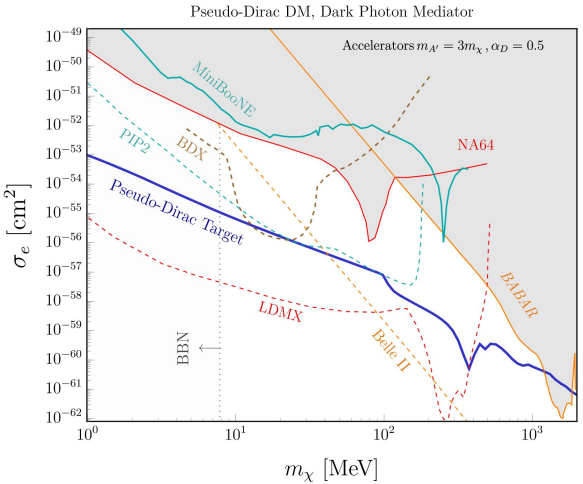
Changing spin of the DM can open more parameter space.



$$\frac{1}{2}\bar{\chi}\gamma^\mu\gamma^5\chi$$



$$i\chi^*\partial^\mu\chi + \text{c.c.}$$



$$i\bar{\chi}_1\gamma^\mu\chi_2$$

Krnjaic 2025

A Minimal Dark Matter Scenario
Thermalized Secluded Freeze Out

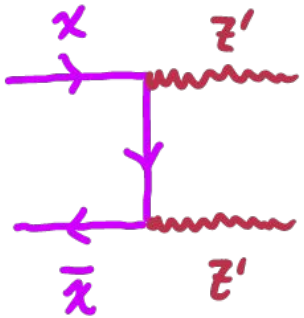
$$m_\chi \gtrsim 1\,\mathrm{GeV}$$
$$m'_Z < m_\chi$$

When **Z'** is lighter than **χ** , the secluded channel dominates

$$\langle \sigma v \rangle_{\chi\bar{\chi}\to Z'Z'} \simeq \frac{\pi\alpha_D^2}{m_\chi^2}$$

- The relic abundance is independent of **ε**

$$\alpha_D \sim 10^{-3} \times (m_\chi/1\,\mathrm{GeV})$$

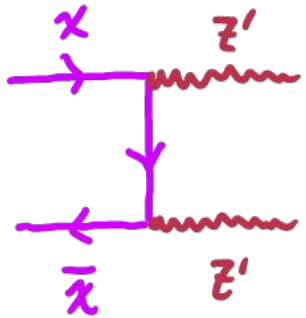


A Minimal Dark Matter Scenario
Thermalized Secluded Freeze Out

$$m_\chi \gtrsim 1\,\mathrm{GeV}$$
$$m'_Z < m_\chi$$

When **Z'** is lighter than **χ** , the secluded channel dominates

$$\langle \sigma v \rangle_{\chi\bar{\chi}\to Z'Z'} \simeq \frac{\pi\alpha_D^2}{m_\chi^2}$$



- The relic abundance is independent of **ε**

$$\alpha_D \sim 10^{-3} \times (m_\chi/1\,\mathrm{GeV})$$

- Thermalization places a lower bound on **ε**

$$\Gamma(Z' \to f\bar{f}),\ \Gamma(Z' f \to \gamma f) \gtrsim H(T_{\mathrm{fo}})$$
$$\epsilon > \epsilon_{\mathrm{therm}} \sim \mathcal{O}(10^{-9} - 10^{-8})$$

A Minimal Dark Matter Scenario
Thermalized Secluded Freeze Out

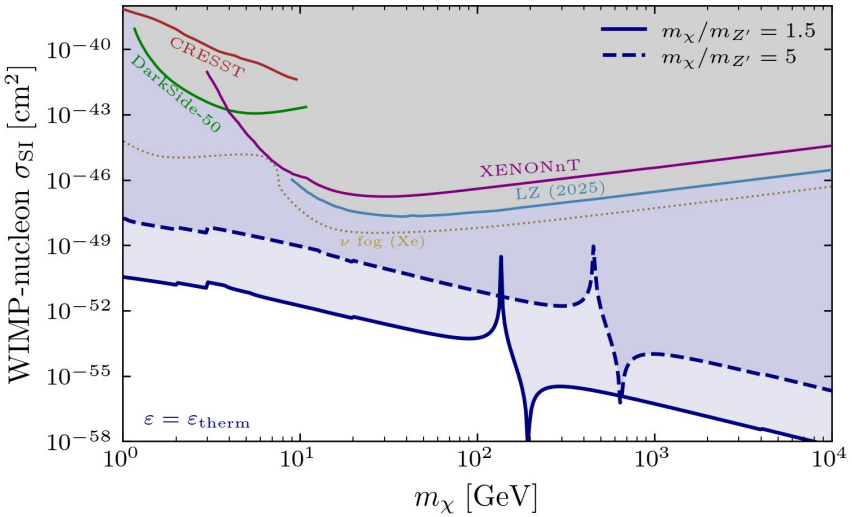
$$m_\chi \gtrsim 1\,\mathrm{GeV}$$

$$m'_Z < m_\chi$$

- No unique direct detection target

$$\epsilon > \epsilon_{\mathrm{therm}} \sim \mathcal{O}(10^{-9} - 10^{-8})$$

- Direct detection experiments are continuously probing this scenario



A Minimal Dark Matter Scenario
Decoupled Secluded Freeze Out

$$m_\chi \gtrsim 1\,\mathrm{GeV}$$

$$m'_Z < m_\chi$$

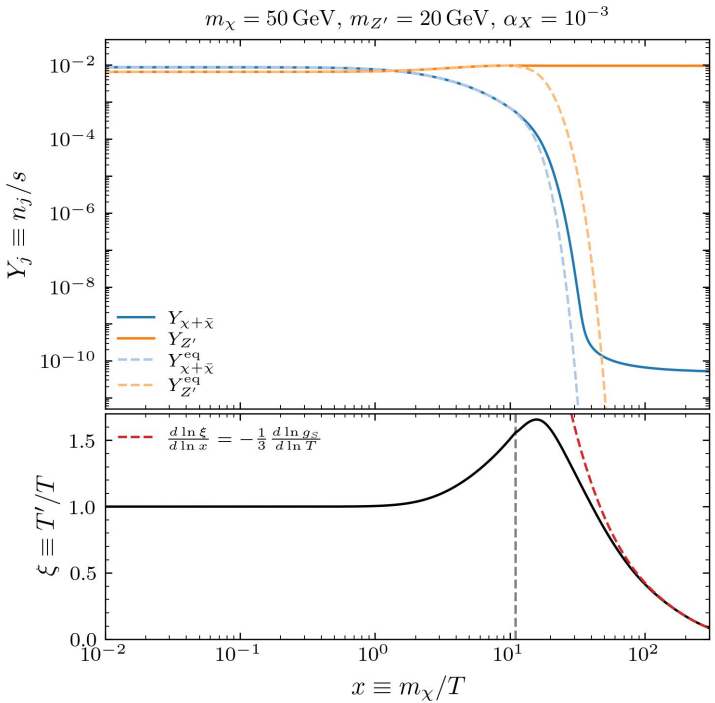
$$\epsilon < \epsilon_{\rm therm}$$

Both sectors begin at the same initial temperature but are now decoupled

- The hidden sector temperature evolves via energy conservation

$$\dot{\rho}_h + 3H\left(\rho_h + P_h\right) = 0$$

- At freeze-out, the sectors have barely diverged so standard **<σv>** is essentially unchanged



Montefalcone, Hooper, Krnjaic 2026 (in prep)

A Minimal Dark Matter Scenario
Decoupled Secluded Freeze Out

$$m_\chi \gtrsim 1\,\mathrm{GeV}$$
$$m'_Z < m_\chi$$

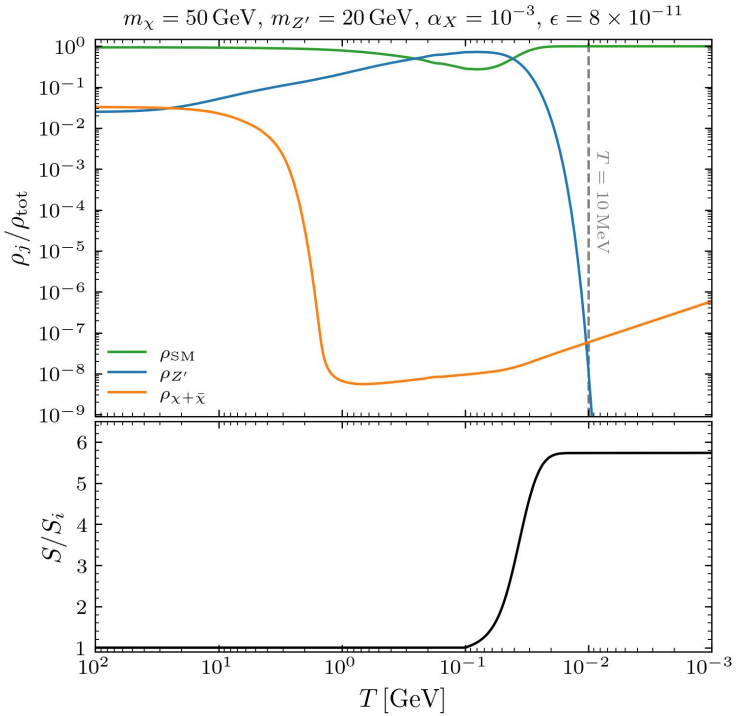
$$\epsilon < \epsilon_{\rm therm}$$

Z' decays injects entropy in the SM bath which dilutes the DM abundance

$$H \sim \Gamma_{Z'} \propto \epsilon^2$$

$$\frac{S_f}{S_i} \approx 20 \left(\frac{10^{-10}}{\epsilon}\right) \left(\frac{m_{Z'}}{100}\right)^{1/2}$$

$$\Omega_c h^2 \approx 0.12 \times \left(\frac{\epsilon}{10^{-11}}\right) \left(\frac{5\times 10^{-4}}{\alpha_D}\right)^2 \left(\frac{m_\chi}{30\,\mathrm{GeV}}\right)^2 \left(\frac{6\,\mathrm{GeV}}{m'_Z}\right)^{1/2}$$



Montefalcone, Hooper, Krnjaic 2026 (in prep)

A Minimal Dark Matter Scenario
Decoupled Secluded Freeze Out

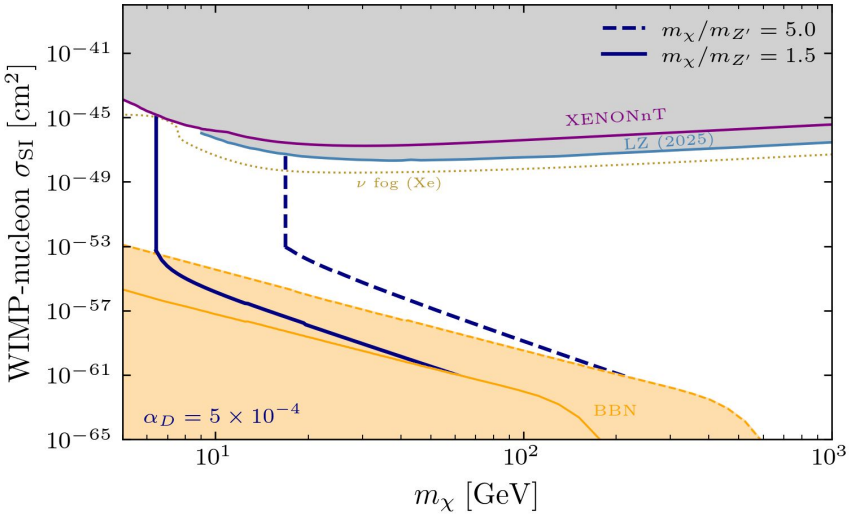
$$m_\chi \gtrsim 1\,\mathrm{GeV}$$
$$m'_Z < m_\chi$$

The dilution mechanism requires:

$$\epsilon \sim 10^{-11} - 10^{-15}$$

- long-lived enough for meaningful entropy injection
- short enough to decay before BBN

$$\sigma^{SI}_{\chi N} \sim 10^{-55} - 10^{-65}\,\mathrm{cm}^2$$



Completely unreachable by any foreseeable experiment

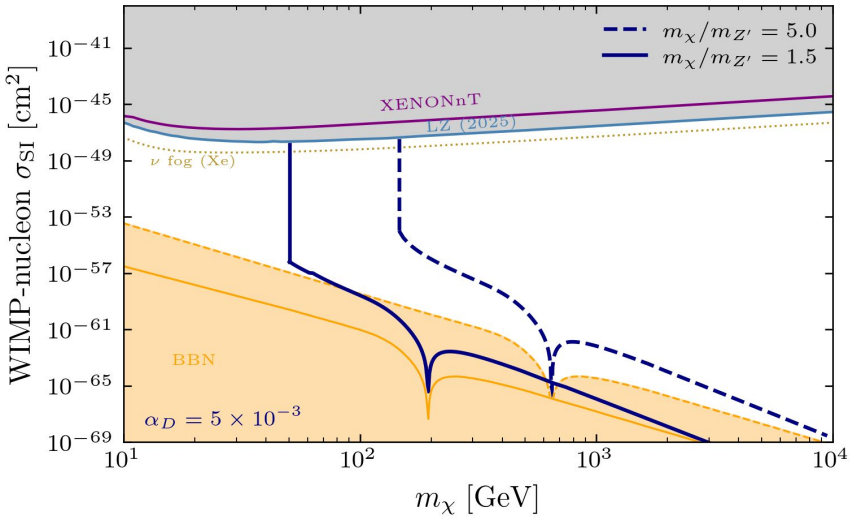
Montefalcone, Hooper, Krnjaic 2026 (in prep)

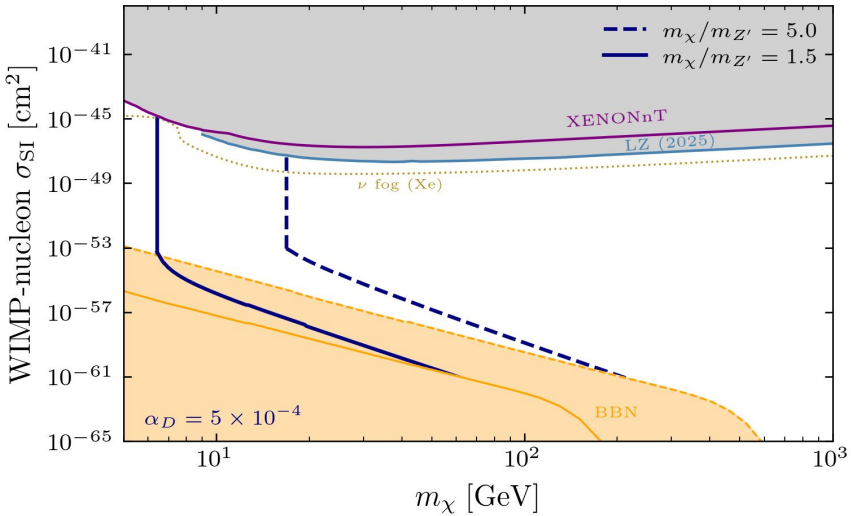
A Minimal Dark Matter Scenario
Decoupled Secluded Freeze Out

$$m_\chi \gtrsim 1\,\mathrm{GeV}$$

$$m'_Z < m_\chi$$

$$\epsilon \sim 10^{-11} - 10^{-15}$$





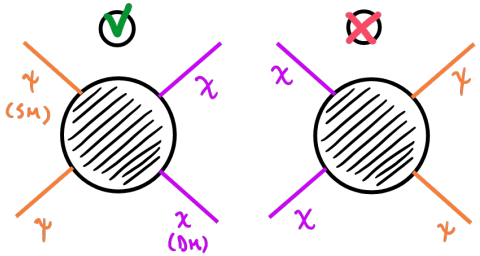
Completely unreachable by any foreseeable experiment

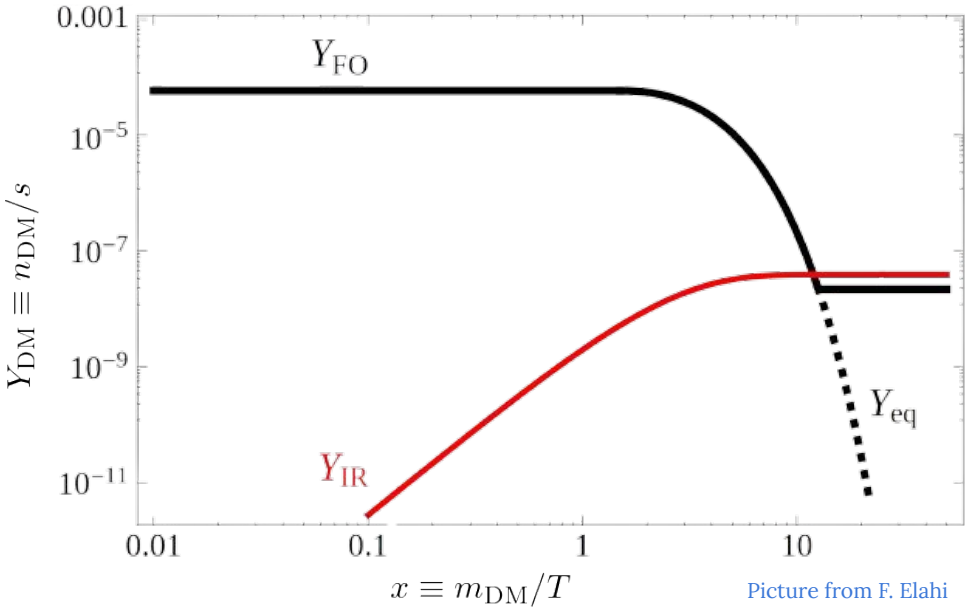
Montefalcone, Hooper, Krnjaic 2026 (in prep)

An alternative Scenario: Freeze-in

Hall, Jedamzik, March-Russell, West 2010

- **Feeble** interaction between DM and the SM:
  DM is never in thermal equilibrium with the SM bath
- Initial DM abundance is negligible





Picture from F. Elahi

DM freezes-in **gradually** and depends on the **lowest T**, i.e. **T□m$_{DM}$**

A Minimal Dark Matter Scenario
Freezing In through a Dark Photon

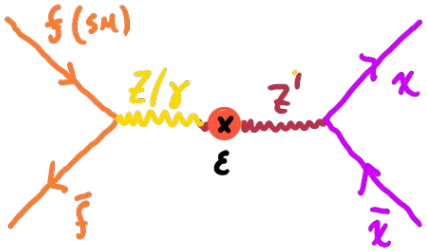
$$1\,\mathrm{MeV} \leq m_\chi \leq 1\,\mathrm{GeV}$$
$$m_{Z'} \lll 1\,\mathrm{eV}$$

Z' is ultralight

- Avoids astro and cosmological bounds
- leads to large enhancement of the direct detection cross section

$$\kappa \equiv \epsilon\sqrt{\alpha_D/\alpha_{\mathrm{EM}}} \approx \mathcal{O}(10^{-10})$$

Freeze-in through a kinetic mixing



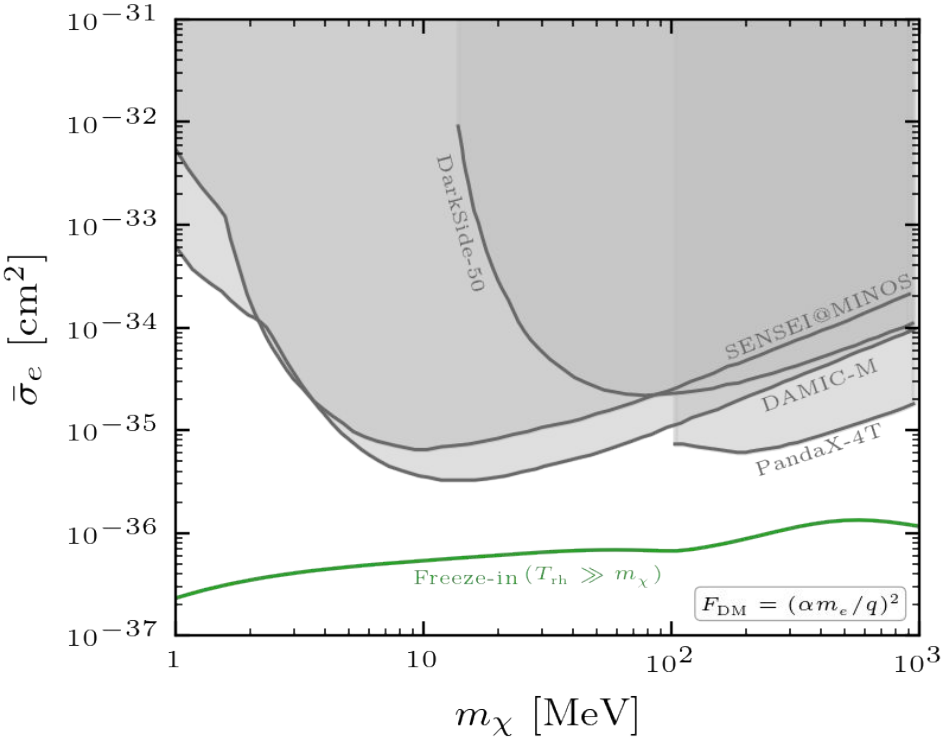
A Minimal Dark Matter Scenario
Freezing In through a Dark Photon

$$1\,\mathrm{MeV} \leq m_\chi \leq 1\,\mathrm{GeV}$$
$$m_{Z'} \lll 1\,\mathrm{eV}$$

$$\overline{\sigma}_e = \frac{16\pi\mu_{\chi e}^2 \alpha^2 \kappa^2}{(\alpha m_e)^4},$$

$$\kappa \equiv \epsilon\sqrt{\alpha_D/\alpha_{\mathrm{EM}}} \approx \mathcal{O}(10^{-10})$$

status as of 03/2025



A Minimal Dark Matter Scenario
Freezing In through a Dark Photon

$$1\,\mathrm{MeV} \leq m_\chi \leq 1\,\mathrm{GeV}$$
$$m_{Z'} \lll 1\,\mathrm{eV}$$

- For $T_{rh} << m_\chi$: $\Gamma_{production} \sim \exp(-2m_\chi/T)$

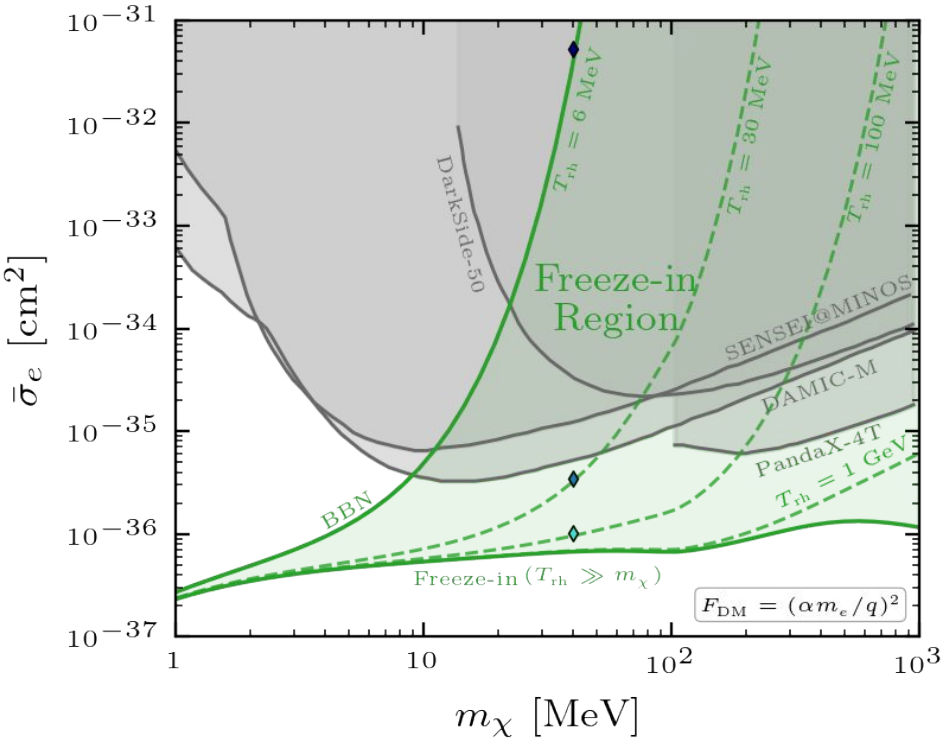
  Kuzmin, Rubakov, 1998; Bringmann et. al. 2021; Cosme et al. 2023;

  - only SM particles in the **tail** of their distributions have energy to annihilate into DM with $m_\chi >> T$

- To counteract the suppressed production, we need **a larger portal coupling**

$$\frac{\kappa(T_{\rm rh} \ll m_\chi)}{\kappa(T_{\rm rh} \gg m_\chi)} \sim \sqrt{x_{\rm rh}} e^{x_{\rm rh}}$$

status as of 03/2025

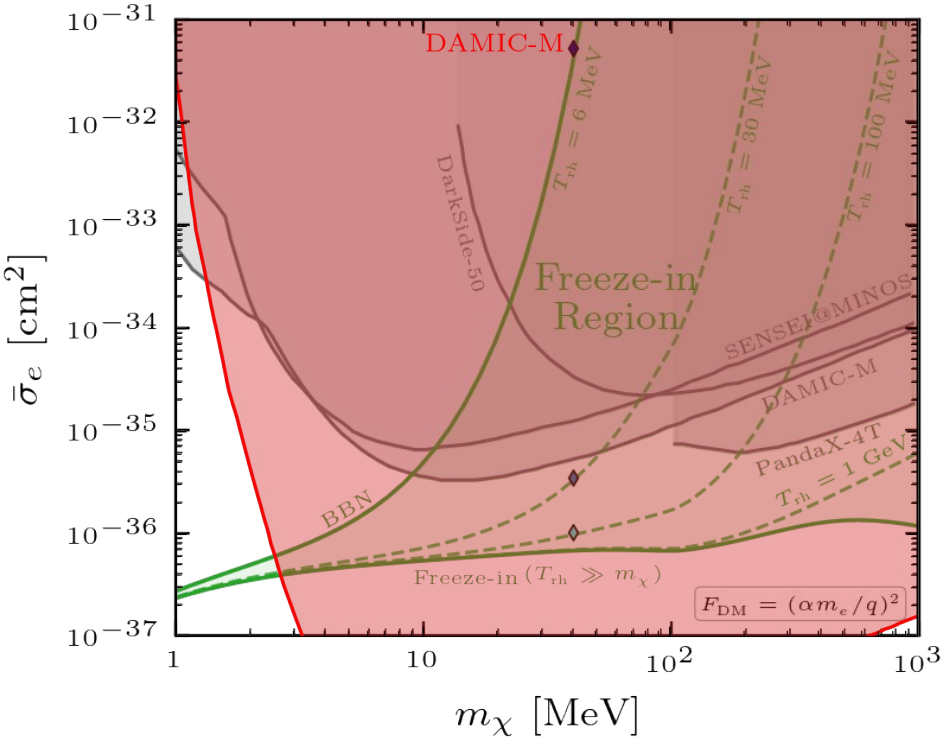


Boddy, Freese, **Montefalcone**, Shams Es Haghi 2024

A Minimal Dark Matter Scenario
Freezing In through a Dark Photon

$$1\,\mathrm{MeV} \leq m_\chi \leq 1\,\mathrm{GeV}$$
$$m_{Z'} \lll 1\,\mathrm{eV}$$

status as of 04/2025



The newest DAMIC-M results effectively exclude all of these scenarios

$$1\,\mathrm{GeV} \leq m_\chi \leq 1\,\mathrm{TeV}$$

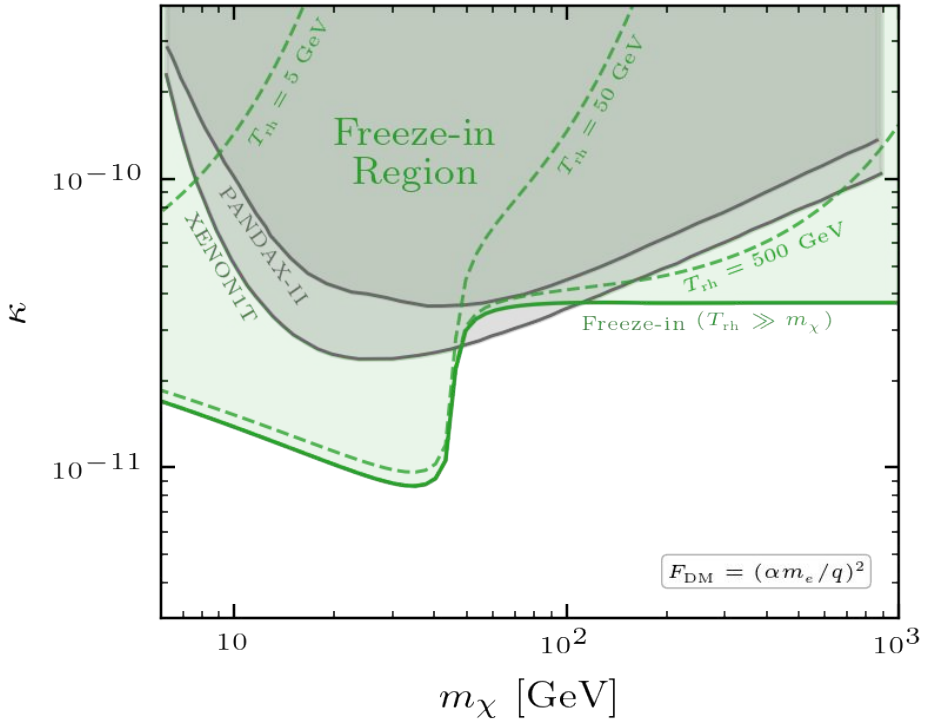
Boddy, Freese, **Montefalcone**, Shams Es Haghi 2024

A Minimal Dark Matter Scenario
Freezing In through a Dark Photon

$$1\,\mathrm{GeV} \leq m_\chi \leq 1\,\mathrm{TeV}$$
$$m_{Z'} \lll 1\,\mathrm{eV}$$

$$\sigma_p = \frac{\mu_{\chi p}^2}{\mu_{\chi e}^2}\bar{\sigma}_e$$

- The same story holds for $\mathbf{m_\chi > 1\ GeV}$
- A large portion of parameter space is currently being **probed by direct detection**



Boddy, Freese, **Montefalcone**, Shams Es Haghi 2024

A Minimal Dark Matter Scenario
More in the Landscape

- Resonance ($m_{Z'} \approx 2m_{DM}$) — enhancement $\sim(m_{Z'}/\Gamma_{Z'})^2$; only surviving window for scalar DM.
- KINDER ($m_{DM} < m_{Z'} < 2m_{DM}$) — forbidden channels, off-shell, and 3 → 2 processes dominate. Fitzpatrick et al. 2022
- Reannihilation / leak-in ($\varepsilon \sim \varepsilon_{therm}$) — neither thermalized nor frozen in; partial thermal contact gives out-of-equilibrium abundances. Alenezi et al.. 2025
- PeV–EeV secluded DM ($m_{DM} > 1$ PeV) — entropy dilution pushes thermal DM to the unitarity limit. Berlin, Hooper, Krnjaic 2016
- Other DM spins — Majorana (*p*-wave, evades CMB), pseudo-Dirac (loop-suppressed direct detection), etc… Krnjaic 2025